



US0D1091822S

(12) **United States Design Patent** (10) **Patent No.:** **US D1,091,822 S**
Van Rensburg et al. (45) **Date of Patent:** **** Sep. 2, 2025**

(54) **FIXATION ARTICLE** 5,672,178 A * 9/1997 Petersen A61B 17/68
606/907
(71) Applicant: **DDS Company, Inc.**, Durham, NC 5,755,574 A * 5/1998 D'Alise A61C 8/0001
(US) 433/173
5,927,979 A * 7/1999 Misch A61C 8/0022
433/172

(72) Inventors: **Cornelis J. Janse Van Rensburg**,
Durham, NC (US); **Matthew Vrhovac**,
Durham, NC (US); **Richard Meaney**,
Durham, NC (US); **Louis Costanzo**,
Durham, NC (US)

(Continued)

FOREIGN PATENT DOCUMENTS

CN 304914063 * 11/2018
CN 305063459 * 3/2019

(Continued)

(73) Assignee: **DDS Company, Inc.**, Durham, NC
(US)

OTHER PUBLICATIONS

(**) Term: **15 Years**

Angulated Multi Unit Abutment, posted Apr. 29, 2025 [online],
[retrieved Apr. 29, 2025]. Retrieved from internet, <https://dipdental.com/products/dsi-premium-angulated-multi-unit-abutment-m1-6-4-8mm-full-set-conical-connection-np-03-5mm> (Year: 2025).*

(Continued)

(21) Appl. No.: **29/796,270**

(22) Filed: **Jun. 23, 2021**

(51) **LOC (15) Cl.** **24-03**

(52) **U.S. Cl.** **D24/155; D8/387**
USPC **D24/155; D8/387**

(58) **Field of Classification Search**
USPC D24/152, 155–157, 170, 171, 176, 147,
D24/180; D8/387

CPC A61B 2017/5164; A61B 17/176; A61B
34/30; A61C 1/084; A61C 8/0089; A61C
8/0022; A61C 3/02; A61C 3/00; A61C
19/02; A61C 13/30; A61C 17/005; B25B
13/18; B25B 13/44; A61M 25/01

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,976,739 A * 12/1990 Duthie, Jr. A61C 8/0022
606/60
5,061,181 A * 10/1991 Niznick A61C 8/0089
433/174
5,078,607 A * 1/1992 Niznick A61C 8/0022
433/173

Primary Examiner — Lilyana Bekic

Assistant Examiner — Anne E. Bedney

(74) *Attorney, Agent, or Firm* — Ward and Smith, P.A.;
Ryan K. Simmons

(57)

CLAIM

The ornamental design for fixation article, as shown and
described.

DESCRIPTION

FIG. 1 is a perspective view of a fixation article;

FIG. 2 is a front view thereof;

FIG. 3 is a rear view thereof;

FIG. 4 is a left side view thereof;

FIG. 5 is a right side view thereof;

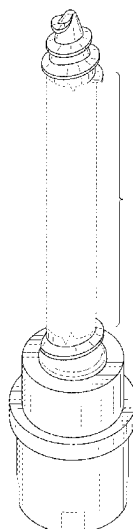
FIG. 6 is a top view thereof;

FIG. 7 is a bottom view thereof; and,

FIG. 8 is a cross-sectional view thereof.

The broken lines in the drawings illustrate portions of the
fixation article that form no part of the claimed design.

1 Claim, 5 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

6,251,112	B1 *	6/2001	Jackson		A61B 17/7091	606/300
D616,548	S *	5/2010	Fares		D24/156	
D629,105	S *	12/2010	Horton		D24/155	
D663,032	S *	7/2012	Mashio		D24/156	
D666,298	S *	8/2012	Sibhatu		D24/156	
D732,669	S *	6/2015	Fisler		D24/156	
D741,494	S *	10/2015	Wagner		D24/176	
D765,862	S *	9/2016	Schendel		D24/156	
D840,037	S *	2/2019	Sarig		D24/155	
D840,038	S *	2/2019	Sarig		D24/155	
D840,039	S *	2/2019	Sarig		D24/155	
D853,562	S *	7/2019	Thomé		D24/152	
11,166,793	B2 *	11/2021	Zipprich		A61C 8/0013	
D952,155	S *	5/2022	Joseph		D24/156	
D956,233	S *	6/2022	Hyeon		D24/155	
D967,427	S *	10/2022	Kim		D24/156	
D993,416	S *	7/2023	Blay		D24/156	
D1,023,308	S *	4/2024	Park		D24/156	
D1,031,043	S *	6/2024	Blay		D24/156	
D1,046,156	S *	10/2024	Lassila		D24/156	
D1,069,121	S *	4/2025	Linder		D8/387	
12,290,289	B1 *	5/2025	Zhang		A61B 17/7208	
12,290,292	B2 *	5/2025	Fauth		A61B 17/17	
12,303,170	B2 *	5/2025	Nichols		A61B 17/7082	
12,303,377	B2 *	5/2025	Gill		A61F 2/0811	
2002/0110784	A1 *	8/2002	Kumar		A61C 8/008	433/172
2002/0172923	A1 *	11/2002	Strong		A61C 8/0089	433/165
2003/0082497	A1 *	5/2003	Kastenbaum		A61C 8/0089	433/144
2004/0260298	A1 *	12/2004	Kaiser		A61F 2/0811	606/232
2005/0026114	A1 *	2/2005	Nilo		A61C 8/0089	433/165
2005/0170311	A1 *	8/2005	Tardieu		A61C 1/084	433/76
2005/0187549	A1 *	8/2005	Jackson		A61B 17/7032	606/270
2009/0130630	A1 *	5/2009	Suttin		A61C 8/009	433/174
2010/0285427	A1 *	11/2010	Hung		A61C 8/0089	433/174
2012/0178043	A1 *	7/2012	Haber		A61C 1/082	433/165
2013/0244206	A1 *	9/2013	Bullis		A61C 8/0089	433/173
2015/0030993	A1 *	1/2015	von Malottki		A61C 8/0066	433/173
2015/0045803	A1 *	2/2015	Deville		A61B 17/17	606/96
2016/0015488	A1 *	1/2016	Miltau		A61C 9/004	433/196
2016/0022388	A1 *	1/2016	Fässberg		A61C 19/02	433/126
2016/0235504	A1 *	8/2016	Siegmund		A61C 8/006	
2018/0125615	A1 *	5/2018	Goodman		A61C 8/0089	
2019/0201169	A1 *	7/2019	Goodman		A61C 8/0089	
2021/0153983	A1 *	5/2021	Soler Brullet		A61C 8/006	
2021/0338375	A1 *	11/2021	Wang		A61B 17/1673	
2022/0409220	A1 *	12/2022	Van Rensburg	...	A61B 17/8625	
2025/0127627	A1 *	4/2025	Bullard		A61F 2/4611	
2025/0129920	A1 *	4/2025	Hong		F21V 23/008	
2025/0152202	A1 *	5/2025	Mullaney		A61B 17/62	
2025/0152203	A1 *	5/2025	Mullaney		A61B 17/6475	
2025/0155059	A1 *	5/2025	Smoke		F16L 23/032	
2025/0160901	A1 *	5/2025	Schlaepfer		A61B 17/7032	

FOREIGN PATENT DOCUMENTS

CN	307434220	*	7/2022
CN	308085694	*	6/2023
CN	308548307	*	3/2024
CN	308964318	*	11/2024
JP	D1477183	*	8/2013
JP	D1729223	*	6/2014
JP	D1500280	*	11/2022

OTHER PUBLICATIONS

Inside the Box—YouTube, posted Apr. 19, 2021 [online], [retrieved May 1, 2025]. Retrieved from internet, https://www.youtube.com/watch?v=QtfyG_oCROY&t=28s (Year: 2021).*

Absolute Dental Services, Inc post—LinkedIn, posted Apr. 22, 2025 [online], [retrieved Apr. 22, 2025]. Retrieved from internet, https://www.linkedin.com/posts/absolute-dental-services_30yearsofabsoluteexcellence-dentalimplants-activity-7310646817023557632-Ojvl (Year: 2025).*

Ritter Multiunit kit, posted Apr. 22, 2025 [online], [retrieved Apr. 22, 2025]. Retrieved from internet, <https://www.navagation.net/materials?pgid=kvsdihuh-8a5c834e-bba1-49cf-ab09-c984880ba43c> (Year: 2025).*

Bone Fixation Screw Kit , posted Dec. 2022 [online], [retrieved Apr. 28, 2025]. Retrieved from internet, <https://vsr.biohorizons.com/GetDocument?DocumentID=37918> (Year: 2022).*

Guided Surgery Accessories Kit , posted Apr. 29, 2025 [online], [retrieved Apr. 29, 2025]. Retrieved from internet, https://www.physicsforceps.com/index.php?route=product/product&product_id=933 (Year: 2025).*

JD Guided Surgery Kit , posted Jul. 15, 2024 [online], [retrieved Apr. 24, 2025]. Retrieved from internet, https://www.jdentaltcare.com/wp-content/uploads/2024/08/JD-Guided-Surgery-Kit_Product-Catalogue.pdf (Year: 2024).*

Kit 2P Fixation Bidet , posted Apr. 29, 2025 [online], [retrieved Apr. 29, 2025]. Retrieved from internet, <https://www.10000articles.shop/kit-2p-fix-bidet> (Year: 2025).*

NobelPearl Abutment Screw Retrieval Kit , posted Apr. 29, 2025 [online], [retrieved Apr. 29, 2025]. Retrieved from internet, <https://store.nobelbiocare.com/us/en/nobelpearl-abutment-screw-retrieval-kit> (Year: 2025).*

Osseodensification Dental Bur Drills, posted Jan. 8, 2023 [online], [retrieved May 1, 2025]. Retrieved from internet, https://www.alibaba.com/product-detail/Osseodensification-Dental-Bur-Drills-Bone-Expanding_1601099211471.html?spm=a2700.7724857.0.0.57dbb%E2%80%A6 (Year: 2023).*

Screw Fixation System, posted Jun. 23, 2023 [online], [retrieved Apr. 29, 2025]. Retrieved from internet, https://www.zimvie.com/content/dam/zimvie-corporate/en/dental/literature/zv1428/ZV1428_Screw_Fixation_System_Ordering_Info_Final.pdf (Year: 2023).*

Stainless steel guide sleeve, posted Jul. 27, 2022 [online], [retrieved May 1, 2025]. Retrieved from internet, https://www.ebay.com/itm/325054998065?_skw=325054998065&itmmeta=01JT6FKTZ9GRMTZNPEK8SQSDSW&hash=item4baec9a631:g:GcoAAOSwOZZiA9JG&itmp=enc%3AAQA KAAAA8FkggFvd1GGDu0w3yXCmi1ecb32b1Jx1IT6lmlLLf15Ni% (Year: 2022).*

Surgical Guide Kit, posted Apr. 29, 2025 [online], [retrieved Apr. 29, 2025]. Retrieved from internet, <https://www.minimaximplant.com.au/implant-system/surgical-guide-kit/> (Year: 2025).*

ZFX Gentek Titanium Pre-Milled Abutment, posted Apr. 23, 2025 [online], [retrieved Apr. 23, 2025]. Retrieved from internet, <https://eservices.zimviedental.com/product/zfx-gentek-titanium-premilled-abutment-blank-certain/01t5f000004rV97AAE> (Year: 2025).*

* cited by examiner

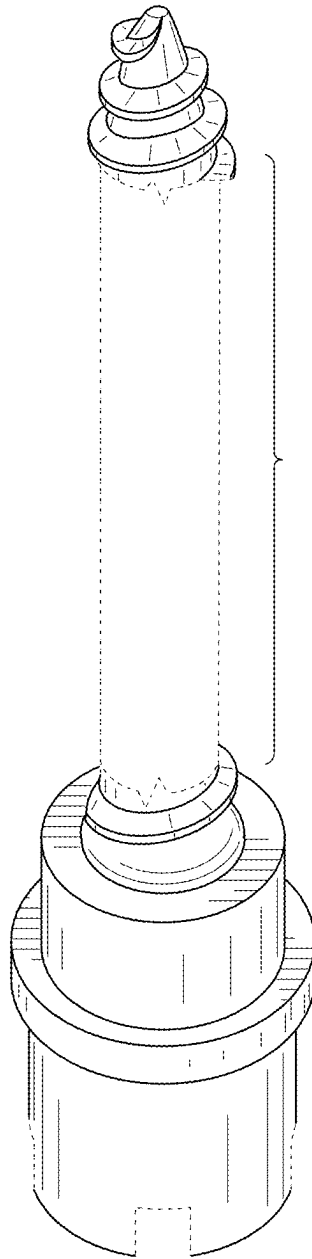


FIG. 1

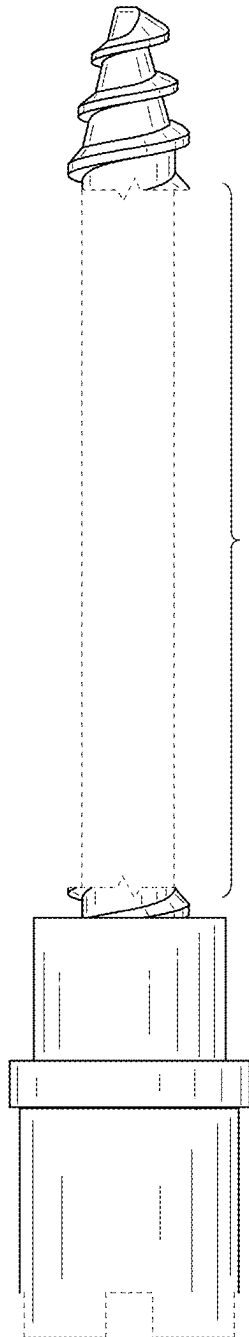


FIG. 2

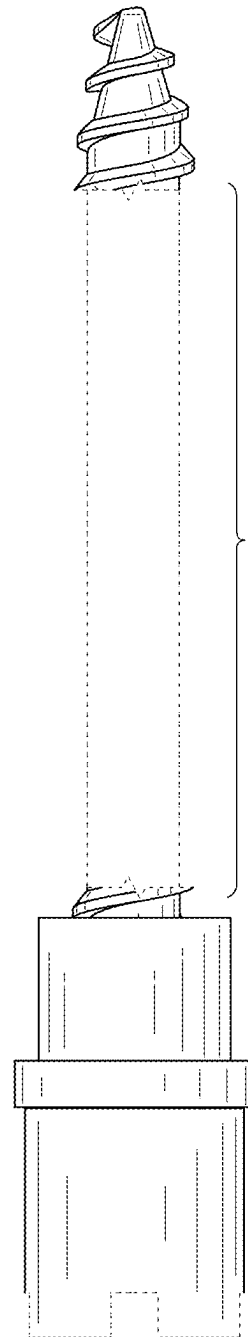


FIG. 3

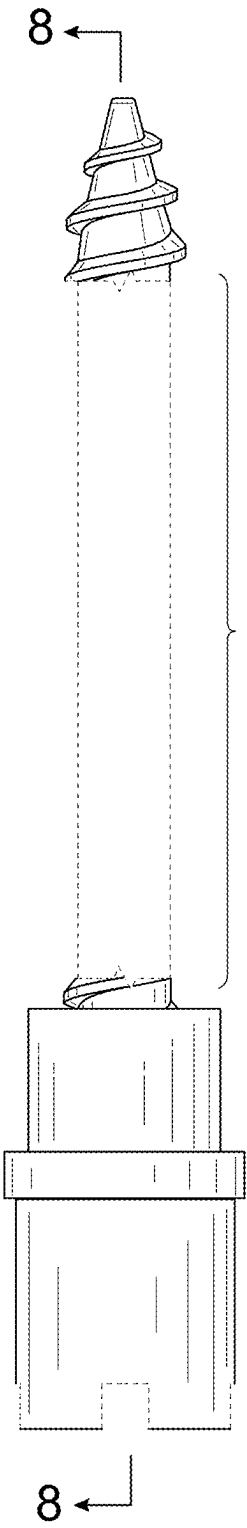


FIG. 4

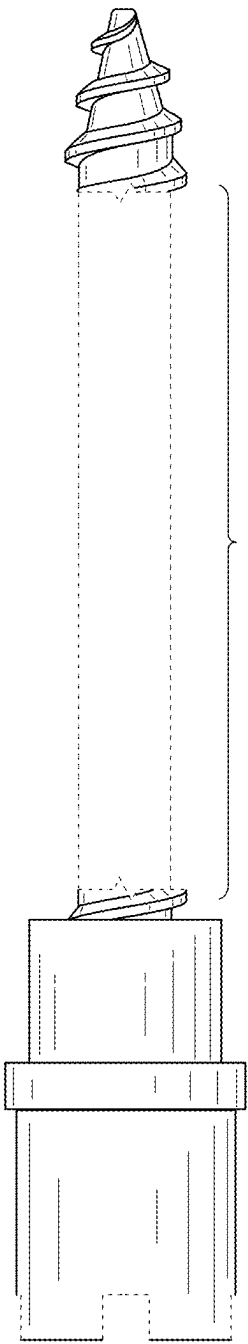


FIG. 5

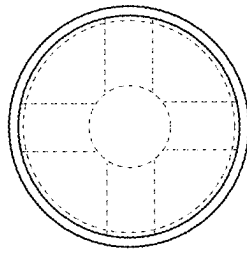


FIG. 6

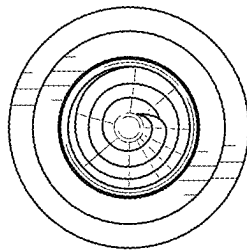


FIG. 7

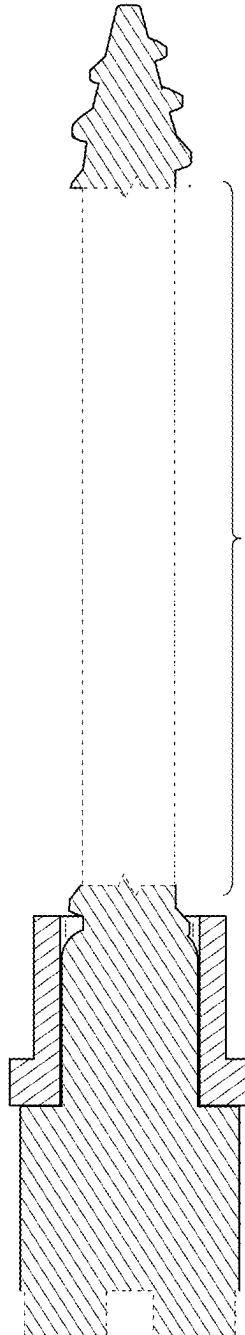


FIG. 8